

**ADMINISTRATIVE INFORMATION****FMI No: 25303A****FMI Title: LightSpeed Ultra Detector Replacement**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	_____
	To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support		
GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked: 2282649, 2280784****FMI Instructions Part No.: 2338095-100 R01****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25303A

FMI Title: LightSpeed Ultra Detector Replacement

Customer Full Name: _____

Number and Street Address: _____

City and Country: _____

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments _____

Effectivity

The following CT Equipment is affected:

Room ID	Model #	Serial #
860545ERULTRA	2282649	276826CN5
262928CT2	2280784	124CD8
475290CT04	2280784	61CD2

Table 1 - Sites Effected

Purpose

The purpose of this FMI is the replacement of defective detectors. These detectors have modules that will have gain drift with large amounts of radiation exposure. This drift may produce an image artifact. FASTCAL will correct for this on a temporary basis until the detector can be replaced.

Prerequisite FMIs

None

Related FMIs

None

Furnished Materials

FMI Kit (2338126) which includes:

Qty	Description	Part #
1	FMI Document	2338095-100
1	Detector	2280784
1	Detector Shipping Container	2227349
1	Detector Replacement Procedure	2335850-100
2	Housing	2255730
4	Screw	2103580-40
4	Clamp & Screw Assembly	2218140-2

Tool Required

Qty	Description	Part #
1	Aero Duster Spray	2226685
1	Amax Contact And Circuit Board Cleaner	2274104
2	Lead blocks (12.7mm X 38.1mm X 88.9mm)	N/A

People Power

Day 1:

8 hours labor (2 engineers)	16 hours total
1 hour travel (2 engineers)	2 hours total

Day 2:

8 hours labor (1 engineer)	8 hours total
1 hour travel (1 engineer)	1 hours total

Detector Damage Test Procedure

Gantry Preparation

1. Lock gantry rotation with the detector at 6 O'Clock position
2. Remove front gantry cover

Phase 1 Record Exam Number

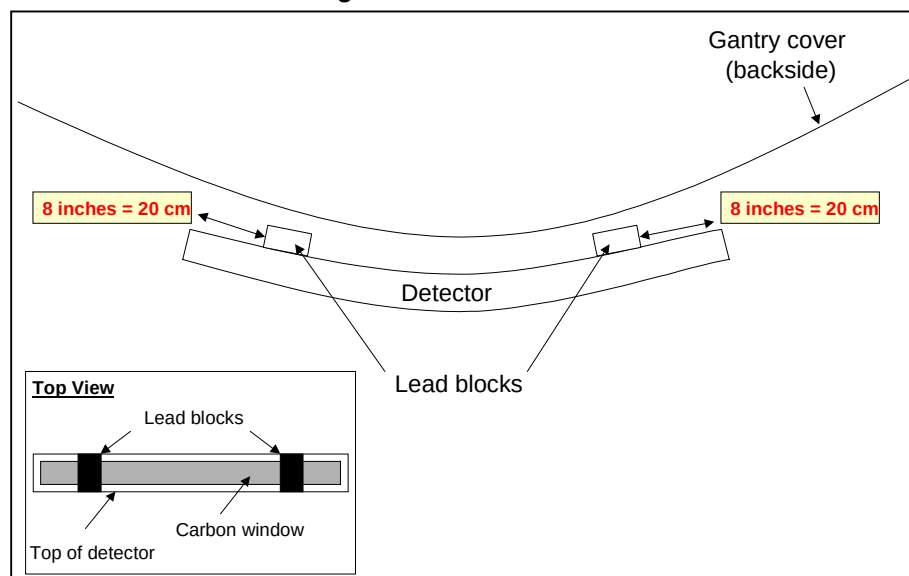
2. Open DDC (Diagnostic Data Collection)
3. Load first protocol (Dw_rad_probe)
4. Execute scans (this will takes 2 scans, write down exam number)

WARNING: NOTE: TIME BETWEEN SCANNING PHASE 2 AND PHASE 3 MUST NOT EXCEED 60 SECONDS!

Phase 2 Record Exam Number

1. Place lead pieces (thickness = 12.7 mm, width = 38.1 mm, length = 88.9 mm) on detector as shown in Figure 1

Figure 1 Lead Placement



2. Load second protocol (Dw_rad_dp)
3. Execute scans (this will take 21 scans, write down exam number)

Phase 3 Record Exam Number

1. When scans are done, quickly remove lead pieces from the detector
2. Load third protocol (Dw_rad_pprobe)
3. Execute scans (this will take 2 scans, write down exam number) --> these scans have to be taken within 60 seconds of the scans with the lead

System Closure and Calibration

1. Re-install gantry cover
2. Due to the high X-ray dose it is recommended the site perform two fastcals (one fastcal only does half of the techniques, two will run all techniques) before they scan the next patient and then two fastcals again 24 hours after that. After that they can return to their normal cal routine.

Data Processing and Archiving

1. Reserve the scan data for each phase (see operators instructions if necessary)
2. Archive Scan Data to MOD (see operators instructions if necessary)
3. Both scans for the pre-damage exam (phase 1)
4. Only scan numbers 2 - 6 of the damage scans (second exam with lead pieces - phase 2)
5. Both scans for the post-damage exam (third exam - phase 3)

Process the data using "Scan Analysis"

1. Start "Scan Analysis"
2. Select the pre-damage exam, first series, first scan (phase 1)
 - 2.1. Select [Create MSD DD File]
 - 2.1.1. select only offset correction
3. Same for 2nd scan of pre-damage scans
4. Same for scan 2 - 6 of the damage scans (phase 2)
5. Same for both of the post-damage scans (phase 3)

Send the MOD via Federal Express to Harrie Netel.

GE Medical Systems
3000 N. Grandview Blvd.
Waukesha, WI. 53188
USA
Attn: Harrie Netel (W-1180)

Harrie Netel will inform you if any further work must be performed.

Detector Replacement Procedure

Preperation or Before you Begin

It is critical that all materials be inspected for damage and availability before starting the detector change. This will minimize the impact of missing or damaged parts on the total process.

It is also very important to verify that **ALL** ESD protection equipment and materials be on-site and in use to prevent damage to the detector and DAS.

1. Verify arrival of replacement detector
2. Inspect replacement detector shipping container for all materials as listed in shipping documents inclosed with the detector. Report any damage to items to engineering immediately.
3. Assure that ESD supplies are available and you understand their use. See LightSpeed Family Detector Change Procedure (2335850-100) Section 1.0 ESD Management Process for details.

Detector Replacement Procedure

Follow the steps listed below:

1. Replace the detector per Direction 2335850-100.
2. Align and calibrate system per Direction 2335850-100.
3. Perform system tests per Direction 2335850-100.

Detector Replacement Procedure

Follow the steps listed below:

1. Repackage old detector and tools in shipping container and return them to:
GE Medical Systems
CT Detector Engineering
3000 North Grandview Blvd.
Waukesha, WI 53189
2. Notify Tim Watkins via e-mail (tim.watkins@med.ge.com) of the shipping information.

Finishing Up

Update the product locator information for your site.

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